ULTASONIC PROXIMITY RADAR SYSTEM

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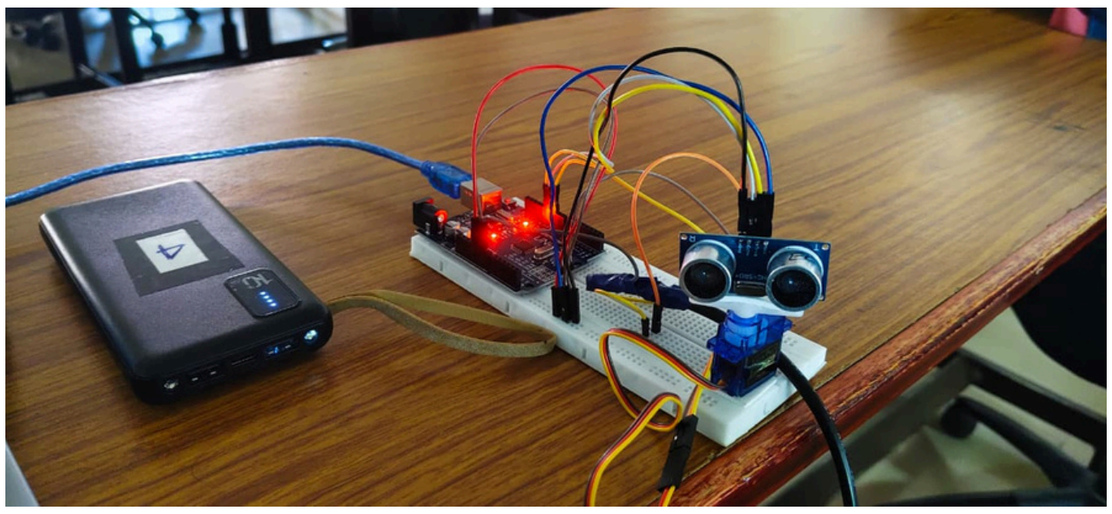
AIM:

Design, prototype, and test an ultrasonic radar sensor system for distance measurement and object detection.

OBJECTIVE:

- To design and implement a non-contact object detection system using an ultrasonic sensor.
- To continuously monitor the detection zone and identify changes when an object enters or leaves the area.
- To display distance readings or object detection status on a serial monitor.
- To develop a reliable and low-cost sensing system suitable for robotics, automation, and security applications.



COMPONENTS REQUIRED

- Ultrasonic Sensor (HC-SR04)
- Microcontroller (Arduino Uno/Nano)
- Servo Motor (SG90)
- Jumper wires, power supply, glue/tape

PROTOTYPE TESTING

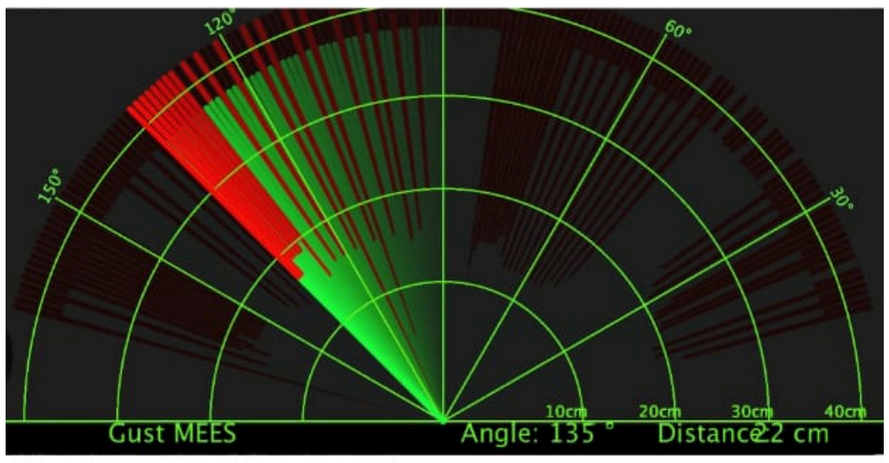
- The prototype was tested indoors under controlled conditions.
- The servo motor rotated the ultrasonic sensor from 0° to 180°.
- Distance measurements were recorded.
- Measured the same distance multiple times to see if the output stayed consistent.
- Tested how the system responded when objects were placed in the detection zone.

METHODOLOGY

- Identify the sensor specifications and required components.
- Assemble the ultrasonic sensor, microcontroller, servo motor, and wiring.
- Write Arduino code to rotate the sensor and record distance readings.
- Visualize the radar sweep using serial plotting or LCD display.
- Conduct multiple tests to measure accuracy and maximum range.

RESULT

- Radar scanning system successfully designed and tested.
- Ultrasonic sensor provided reasonable accuracy for hobby-level radar.
- Results show limitations in range and object type detection.
- Graphical UI clearly visualises environmental mapping.
- System suitable for basic robotics and obstacle detection.



Conclusion

The ultrasonic sensor system proved to be a reliable and efficient setup for real-time distance measurement and obstacle detection. Throughout the project, the prototype delivered stable and accurate readings once it was properly calibrated and tested with different objects, angles, and materials. The system showed that this type of detection is affordable, easy to implement, and highly practical for use in automation, robotics, and safety-related applications where quick and consistent distance feedback is needed. By testing its performance in both controlled conditions and real-life scenarios, it became clear that the setup can adapt well to environmental changes while maintaining accuracy. Overall, the project achieved its objectives and demonstrated how a simple Arduino-based detection system can be developed into a functional and dependable solution.

ACKNOWLEDGMENT
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